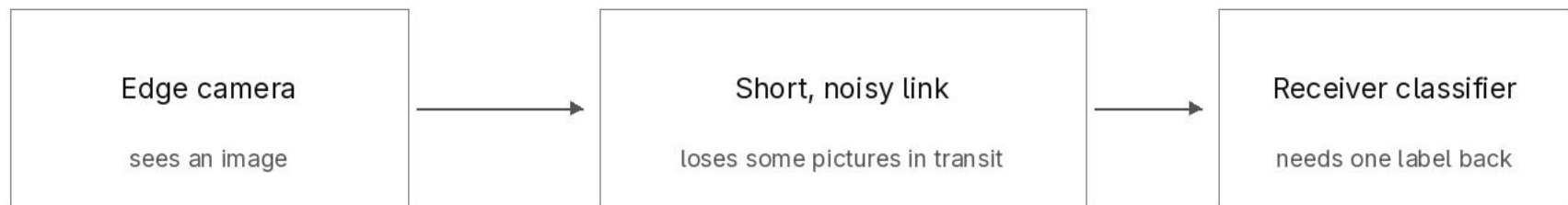


Task-Oriented Deep Joint Source–Channel Coding

What a camera should send when the receiver only needs the answer



Train the sender and the receiver together so the classification decision survives a short, noisy link.

The thesis

Second Review, 29 September – 3 October 2026.

What has changed since the first review

In August the digital system was working. The learned models are trained now.

Aug · Digital system	Measured how badly it loses pictures at every setting. All 288,000 held-back combinations tested, and the settings it should use at each signal strength frozen.
Aug · Learned system	Training loop built, models trained at several seeds. Three of them are frozen at half rate.
Sep · Fair comparison	Built a control that sends learned features digitally, so the comparison includes a task-aware system alongside a compressed image.
Sep · Weak-link training	A second version, with the training signal strength picked at random each time, for comparison against the first.
Sep · Crossing question	Decided it on held-back pictures before seeing the answer: the two lines cross between -5 and -4 dB.
Sep · Power limits	Trained one model under a hard limit on how tall the signal spikes may get.
Oct · Measurements	252 measurements across twelve versions of the system and twenty-one signal strengths, 1,000 held-back pictures each.
Nov · Exam set	Opens in week 11, once the locking document is signed.

Objectives, stated as completion criteria

Five things we set out to build. Each one is checkable.

What we set out to do

- | | |
|----------------------------|--|
| 1. Build it | Train a learned sender and receiver together, so the picture goes through the same kind of noise a radio adds. |
| 2. Same radio time | At every signal strength and rate, both systems get the same number of chances to use the radio. |
| 3. Charge the overhead | Checksums, padding and error-correction bytes come out of the budget before any picture data. |
| 4. Pick the setting fairly | The operating point is chosen using the digital system alone, before anyone sees the learned curve. |
| 5. Report per picture | Every row records what happened to one picture, including those that failed to arrive. |

How these are scored

The rubric scores whether this list was completed.

Completion is independent of which way the curve went. A crossing and the learned system winning everywhere both finish the list on the left.

Before the numbers, one picture

Every chart in this deck runs from a very weak radio signal on the left to a clean one on the right.

Plain words	Our term	Where you see it
How strong the signal is	SNR, in decibels	the left-to-right axis
How many pictures got through	delivery coverage	the flat 10% line
How many we labelled correctly	top-1 accuracy	the up-and-down axis
How many times we use the radio	channel-use budget	12,800 or 3,200 per picture

Every point on every chart is the same 1,000 pictures, at that one signal strength.

One example

At the weakest signal we measured:

The learned system labelled 728 of the 1,000 pictures correctly.

The digital system got almost nothing through, so every picture fell back to one fixed answer. That answer happens to be right 100 times out of 1,000. So 10%.

That flat 10% is the fallback rule: 1,000 pictures that never arrived.

What we measured for this review

We built twelve versions of the system and tested each one at twenty-one signal strengths.

Item	Figure	What it is
Measurements	252	One row each in the published record
Versions of the system	12	Five are controls, not headline systems
Signal strengths	21	From very weak to very clean
Pictures per point	1,000	The same held-back pictures every time
Exam pictures opened	0	Sealed until week 11

Two things to keep in mind

Radio time Half rate uses the radio 12,800 times per picture; quarter rate uses it 3,200 times. Within one rate both systems get exactly the same amount of radio time.

Who marks it The learned systems are marked by the classifier built into them. The digital systems are marked by one we trained on compressed-looking images. So every comparison in this deck is between whole systems, link and marker included.

The two models behind every number

One classifier reads the pictures. One learned link decides what to transmit.

The learned link

Part	What it is
Encoder	Conv stack, 64 then 128 channels
Blocks	2 residual blocks, Group norm, PReLU
Decoder	Mirror of the encoder, two heads
Heads	Class scores, plus a rebuilt picture
Width sent	8 complex symbols per use at half rate
Parameters	1.64 M at the widest rate

How it was trained

Setting	Value
Objective	Class scores + $3.0 \times$ reconstruction
Optimiser	Adam, learning rate 0.001
Schedule	Cosine decay, no warm-up
Batch, epochs	32 pictures, 100 epochs
Augmentation	Random crop, horizontal flip
Precision	Mixed 16-bit

The classifier

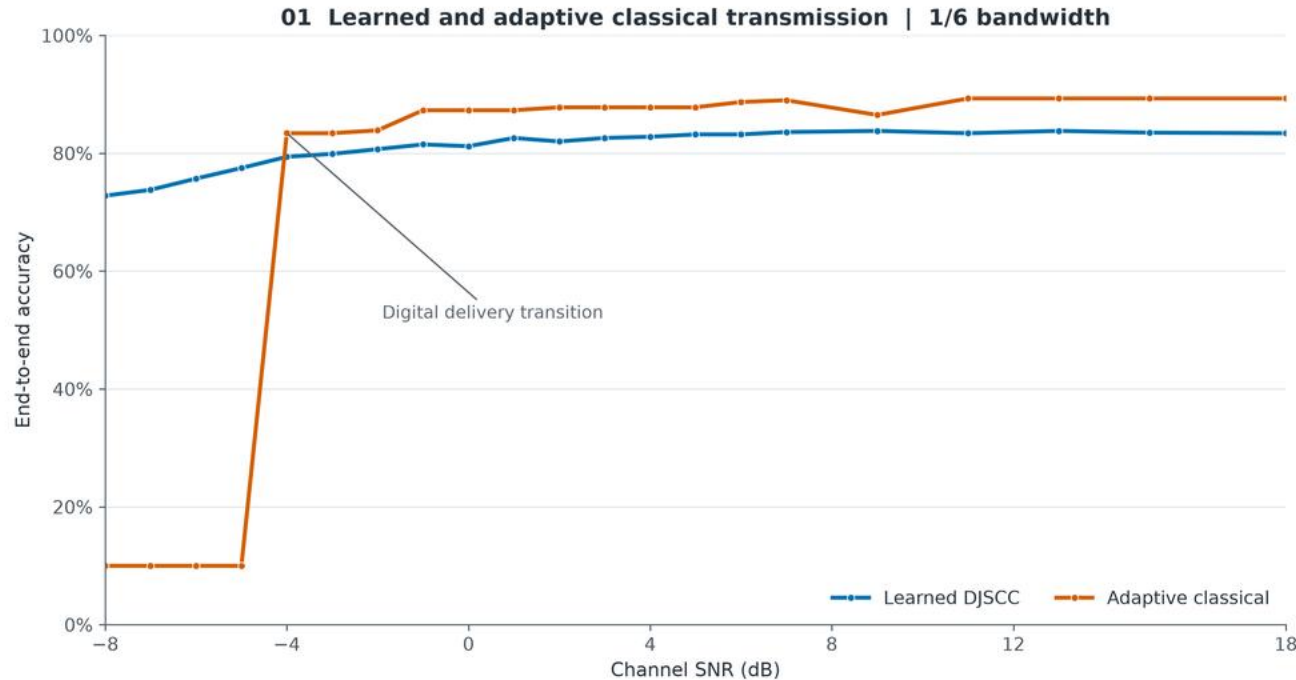
ResNet-18, 11.18 M parameters, trained from scratch. It reads 89.8% of held-back pictures correctly, and that is the ceiling both systems are measured against.

Training cost

48.7 seconds per epoch and 1.0 GiB of graphics memory, measured on a laptop GPU.

Where the digital system gives up, and where it takes over

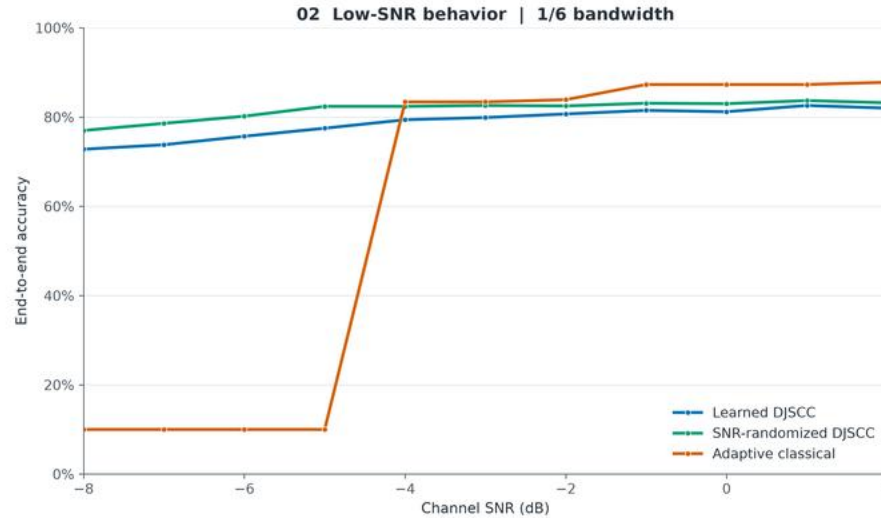
One line runs the whole width of the chart. The other has a step in it near the left.



At the weakest signal the learned system is right on 728 of 1,000 while the digital system delivers nothing. From -4 dB it delivers and goes ahead. At +18 dB it is well ahead.

Close-up: the moment the digital system starts working

Between -5 and -4 dB the digital system begins getting pictures through, and its score jumps 73 points.



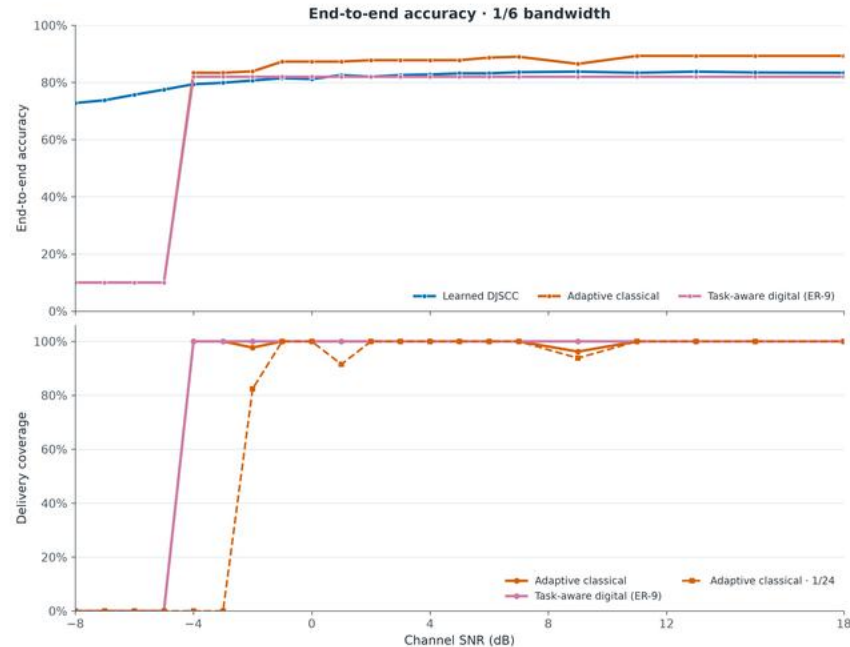
At -8 dB The learned system is right on 728 of 1,000. The digital system got nothing through, so it scores 10%.

At -4 dB The digital system now gets everything through and reaches 83.4%, ahead of the learned system's 79.4%.

After that It dips at -2 dB and again at $+9$ dB.

Why the digital line breaks where it does

The score only moves when the pictures start or stop arriving.



The 10% is a rule

Every picture takes the same fallback answer when delivery fails. On an evenly split set that is right 100 times in 1,000.

Outages are part of it

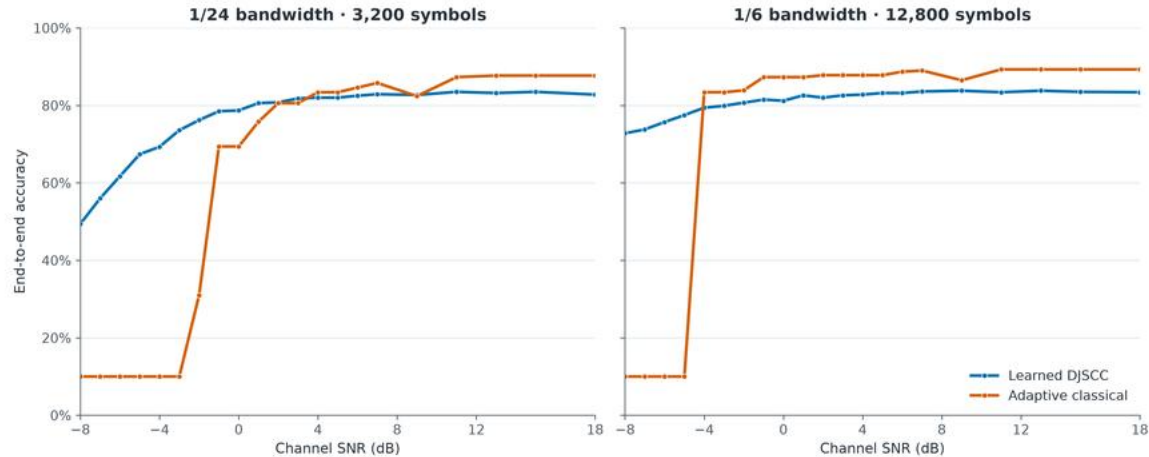
We report how many pictures arrived next to how many we got right, so a broken link is distinguishable from a wrong guess.

Learned always answers

Its rules promise a label at every strength, so its curve never flattens against a floor.

What happens when we quarter the radio time

On a quarter of the radio time, the digital system loses far more at the weak end than the learned one does.



Weak signal

Going from half to quarter rate drops the learned system from 72.8% to 49.4%.

At 0 dB, quarter rate

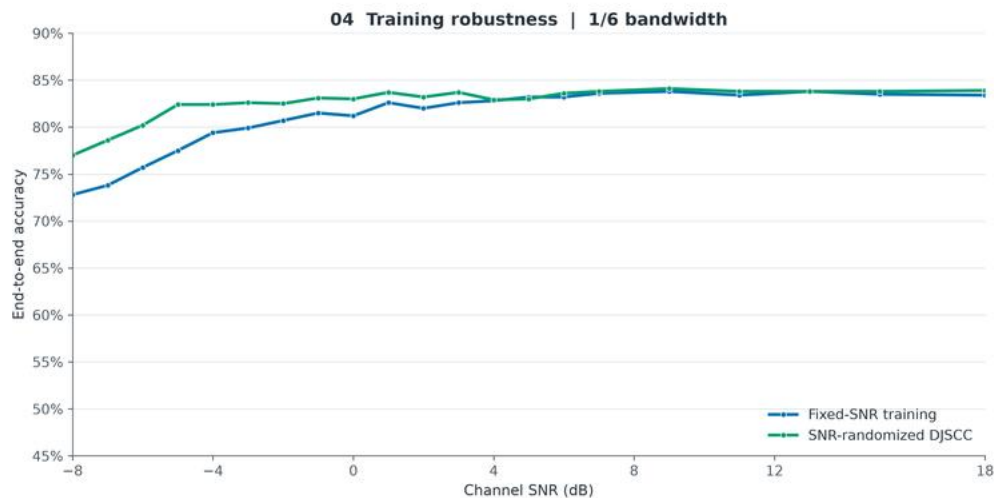
Learned 78.7% against digital 69.4%. The learned system is still ahead.

Catches up, then slips

Digital passes learned at the +4 dB point, loses the lead at +9 dB, then goes ahead again.

Training across signal strengths instead of one

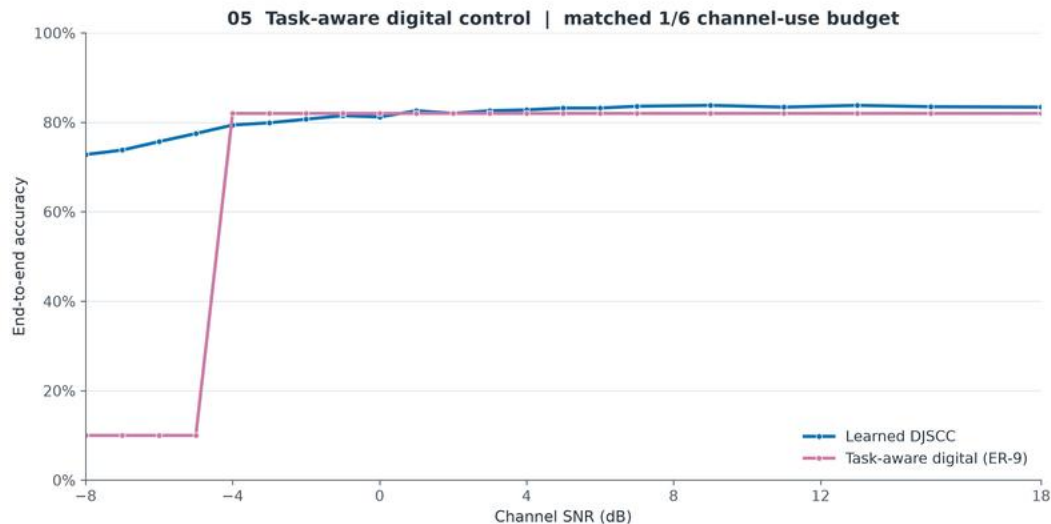
Picking the training signal strength at random each time lifts the whole weak end of the line.



Training	-8 dB	-4 dB	0 dB	+9 dB	+18 dB
Random each time	77.0	82.4	83.0	84.1	83.9
One fixed strength	72.8	79.4	81.2	83.8	83.4

The control that keeps the comparison honest

A JPEG image is an easy thing to beat. This system is harder, and it is the one worth reporting.

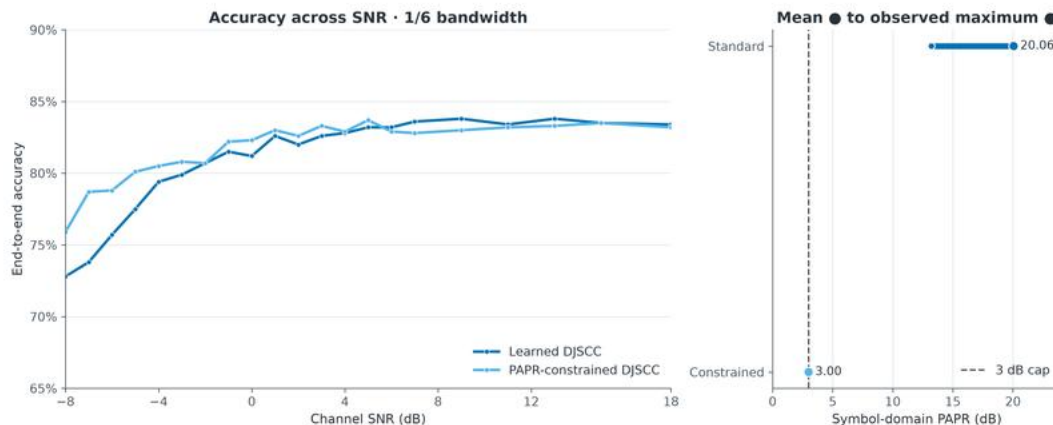


What this system does Instead of a JPEG image it sends the learned features themselves, through the same error-correction code and radio settings, using the same 12,800 uses of the radio. It separates "knows the task" from "codes both at once".

What it settles It holds 82.0% at every strength from -4 dB up. Sending learned features digitally is strong once pictures get through, so the weak-signal gap is about failed deliveries as much as representation.

Keeping the signal peaks from getting too tall

A real transmitter needs a limit on signal peaks, so we trained a second model under one.

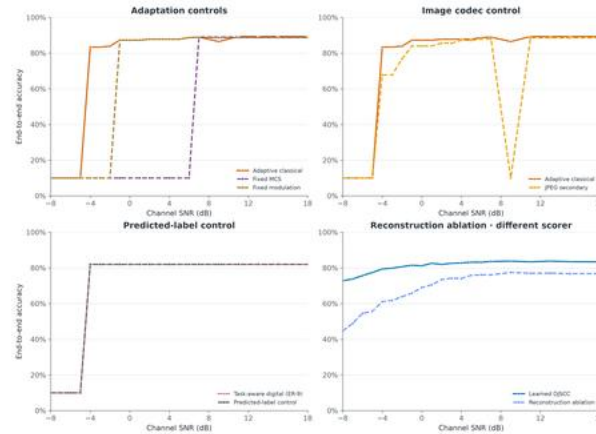


“Peak-to-average power ratio” = how much taller the tallest spike is than the average.

Model	Tallest spike	At -8 dB	At +18 dB
Power-limited	3.000002 dB	75.9%	83.2%
Ordinary learned	20.059284 dB	72.8%	83.4%

Four controls on the digital system

Each panel takes one design choice away and shows what it was worth.



Fixed settings

Sits at the fallback floor until +6 dB, then reaches 89.0%. Choosing settings per strength buys real reach.

Fixed modulation

Starts working later than adaptive selection, and settles near 88.7%.

JPEG secondary curve

88.6% at +18 dB, with a full outage at +9 dB.

Sending the answer

82.0% once it arrives, using the sender's own predicted label.

The numbers behind the curves

How many of the 1,000 held-back pictures we labelled correctly, one picture at a time.

Half rate — 12,800 uses of the radio

System	-8	-5	-4	0	+9	+18
Learned, one strength	72.8	77.5	79.4	81.2	83.8	83.4
Digital, picks settings	10.0	10.0	83.4	87.3	86.5	89.3
Learned, random	77.0	82.4	82.4	83.0	84.1	83.9
Digital, learned feats	10.0	10.0	82.0	82.0	82.0	82.0
Learned, power-limited	75.9	80.1	80.5	82.3	83.0	83.2

Quarter rate — 3,200 uses of the radio

System	-8	-4	-2	0	+4	+18
Learned, one strength	49.4	69.3	76.2	78.7	82.0	82.8
Digital, picks settings	10.0	10.0	31.0	69.4	83.4	87.7

Where the two lines cross

At half rate, which system is ahead flips between -5 dB and -4 dB. We decided the crossing question on exactly that, before seeing the answer.

Reading the two tables

Each column is one signal strength; each cell is the number of correct labels out of the same 1,000 held-back pictures. At half rate the learned row leads up to -5 dB and the digital row leads from -4 dB. At quarter rate the digital row stays on the fallback floor four dB longer, then climbs past the learned row from +4 dB. The difference between the two tables is the effect of cutting radio time fourfold.

How to read these results

One training run	A single observed measurement per figure, from one run over 1,000 held-back pictures.
Whole systems	Marked by their own classifier on one side, by one trained on compressed images on the other. These numbers describe whole systems.
Different amounts of work	Digital settings and grader were frozen from earlier validation work; the learned versions are separate training runs.
Simulated channel	Spike heights are measured on the numbers we send. Real-radio replay is scheduled for late October.
Frozen operating points	The fallback rule and fixed settings produce visible jumps. Every curve passes through measured points.
Both outcomes count as a successful project: a crossing, and the learned system winning everywhere.	

Four times we changed the approach because of what we found

Four changes we made after the measurements started coming in.

Baseline too weak

We had pinned the digital system to the weakest radio setting, which made the lines crossing impossible by arithmetic. We let it pick a stronger setting as the signal improves.

The rule we adopted: every change strengthens the digital system or is decided in advance.

Signal grid too coarse

Our signal strengths were spaced 2 dB apart exactly where the sharpest drop happens, so we could have missed it entirely. We added three more strengths before it mattered.

Put the detail where the effect is.

A faulty check

The script that checks our packet sizes reported no problems while breaking four of its own rules. We fixed the script, and every result came out the same.

All 215 configurations stayed feasible, and every scientific outcome was unchanged.

Exam set unlocked too early

The exam set was set to unlock in week 9, about three weeks before the document that locks it existed. We moved that to week 11.

They have stayed locked ever since.

Twenty-two further changes were made during the project, including three checks that turned out to be wrong and were repaired.

What we claim as new

The task-aware digital control

A second system sends learned features through the same error-correction code and radio settings, using the same amount of radio time. It tells “knows the task” apart from “codes both at once” inside one experiment.

Prior work reports the gap without this control.

Charging the overhead to the digital system

The picture file, framing, checksums and rate-matching bytes are all measured and counted, so the digital system is tuned on the payload it can genuinely use.

Prior work quotes a nominal rate and leaves the overhead uncounted.

What this builds on

- Getting steadily worse as the link worsens
- Avoiding the sudden drop-off
- Sending task-aware features digitally
- Training a neural encoder and decoder

All four are established in the prior literature. This project builds on them.

Negative check

Each claim resolves to a specific entry in the 30-source literature review. The full written novelty statement ships with the final review package.

Where the remaining weeks went

29 Sep – 3 Oct	Second Review	Held-back measurements complete; package and figures frozen	done
5 – 11 Oct	Open the exam set	One guarded opening once the freeze is signed	planned
12 – 18 Oct	Decide everything	Every hypothesis settled on the exam set; first tier frozen	planned
19 – 25 Oct	Demo	Signal-strength slider, both systems side by side, frozen plot	planned
26 Oct – 1 Nov	Poster and hardware	Poster draft; real radio only if the first tier leaves room	planned
2 – 8 Nov	Report freeze	Report finished in the required format	planned
9 – 15 Nov	Spare week	Report completion and checking	planned
16 – 22 Nov	Final Review	Final review 17–21 Nov; report and material due 20 Nov	planned

The report is due on 20 November, inside Final Review week, so the week before it is a freeze.

Where this leaves the project

The learned system keeps answering when the link is too weak for the digital one to get a single picture through. Once the digital system starts delivering, it is the more accurate of the two. They cross between -5 and -4 dB.

Next One guarded run on the exam set once the locking manifest is signed.

Then Every hypothesis settled on the exam set, and the first tier frozen.

Later Real-radio replay and the live demo run once the first tier is frozen.

What we are asking for

Questions on the method or the numbers. The per-picture records behind every point are available on request, and the exam set stays sealed until week 11.